

Exploring Spectra

Austin Efflandt, Cirdan Klindworth

I. Introduction

The field of spectroscopy was introduced by Joseph von Fraunhofer in 1814 when he invented the first spectroscope to measure the spectrum of light. This concept is very important and interesting for physics and chemistry because objects that emit light when heated can be detected by these spectroscopes, and the position of the lines emitted define which element was emitting the light and could be observed through a telescope. In 1842, Alexandre Edmond Becquerel was the first to record a photograph of the sun's spectrum, which gave way to a more modern digital approach to spectrometers, the spectrograph, which was improved throughout the 20th century into the spectrometers we use today. Spectrometers today use programs to graph the electromagnetic emission spectrum onto a computer using a photodetector.

Our focus for this experiment is to explore the efficiency of the broadband spectrometer, as well as derive an equation to correct any inefficiencies of the detector. For this experiment, we will focus more specifically on Planck distributions. We do this by using a calibration lamp that has a given graph which we can use to calibrate our data to be more reliable and accurate. We also want to measure emission spectra for hydrogen and helium by using their respective gas lamps. This will allow us to assess the Balmer series for hydrogen and see how well our spectrometers can detect emission lines in hydrogen and "hydrogen-like" helium atoms. One experiment from N Bibinov, H Halfmann, P Awakowicz, and K Wiesemann discussed how they calibrated their spectrometer mostly by comparison between a "tungsten strip lamp and well-known rovibrational intensity distributions in the emission spectra of N_0 and N_2 ."¹

¹ <https://iopscience.iop.org/article/10.1088/0957-0233/18/5/019/meta>

II. Procedure

Since there are various experiments we are performing, we have several setups to figure out how to approach. Starting with the calibration lamp, we attempt to determine the “efficiency” of each spectrometer. After allowing the lamp to warm up for 30 minutes, we perform measurements for the broadband spectrometer. Our spectrometer picks up a graphical distribution that resembles a blackbody Planck distribution. It represents this as a “relative intensity” versus “wavelength” graph. (See Data and Analysis). We use the calibration lamp’s user manual to observe what the distribution should look like, and then develop the efficiency of the spectrometer as a function of wavelength, converting units where necessary. This will allow us to fix our spectrometer’s inefficient readings at specific wavelengths for other parts of the experiment.

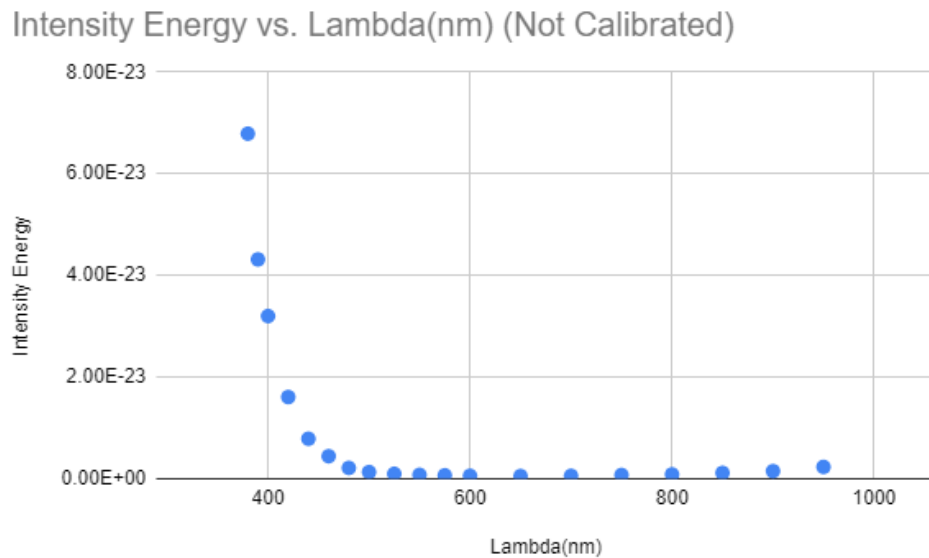
An incandescent light bulb can be made brighter by manually increasing the voltage. Using the broad-range spectrometer, we take measurements while increasing the voltage in 5V increments. Starting at 30V, we work our way up to 125V in order to get a decently large range of filament temperatures. We treat the bulb as a blackbody so that we make sense of our Planck-like graphical readings. As the voltage increases, so does the temperature of the filament. This is the idea we are trying to represent with the Planck function.

Lastly, we turn our attention to the hydrogen and helium lamps. Using both broad and narrow range spectrometers, we take several data for each lamp. The graphical result shows peaks in intensity at certain wavelengths that we interpret as emission spectra for the hydrogen and helium atom. We take the time to find the line width for each emission line as well as an

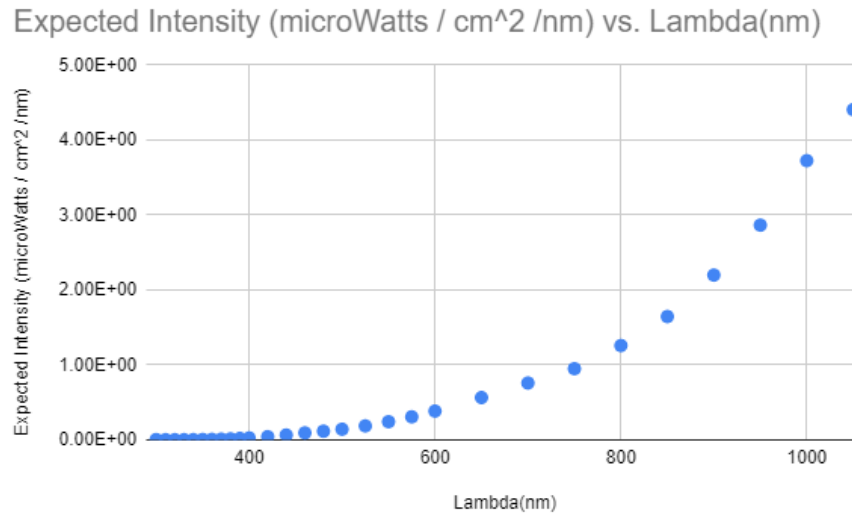
additional method for exploring our spectrometers' resolution. Also, we decide that the narrow range spectrometers do a better job at capturing emission peaks and their associated line widths.

III. Data and Analysis

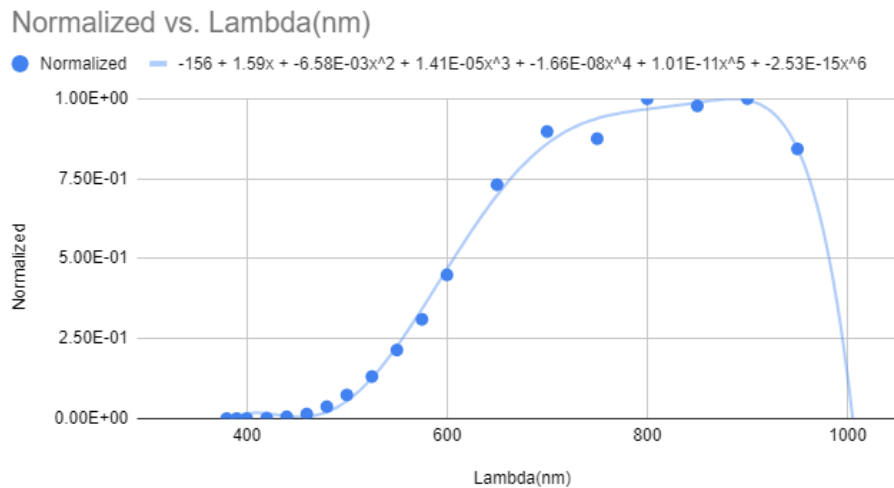
In order to calibrate our data to what is expected from the calibration lamp, we use the manual that came with the calibration lamp to find the expected intensity (in energy) vs. wavelength data. We then take our own data with the same calibration lamp. Note that the intensity measured with our spectrometer has units of “number of photons”, not energy, so we then convert to energy with $E = hc/\lambda$. The figure below is what our data looks like without any calibration.



Compared to the expected data this is very far off, suggesting major inefficiency in the infrared region. Note that the units of the expected slightly vary from our data, but the trend is definitely not accurate without calibration.

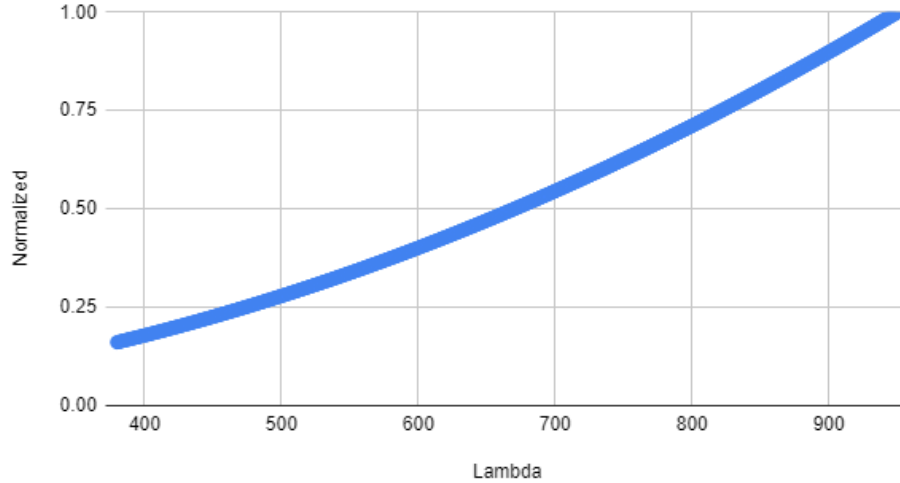


We then compare the expected data to what we measure by dividing each corresponding expected value by the energy we found at that wavelength. This will tell us the efficiency of the spectrometer at each wavelength we measure. We then normalize the compared data and graph it against the wavelength to give us the figure below.



The 6th-degree polynomial fit on the graph is the closest fit we could find to show the trend of the data as it gets more inefficient. With this new line, we can find the calibrated distribution of the energy by plugging in the wavelength for x, yielding the calibrated graph below. Now we can see a trend that is definitely closer to the expected graph seen above.

Normalized vs. Lambda (Calibrated)



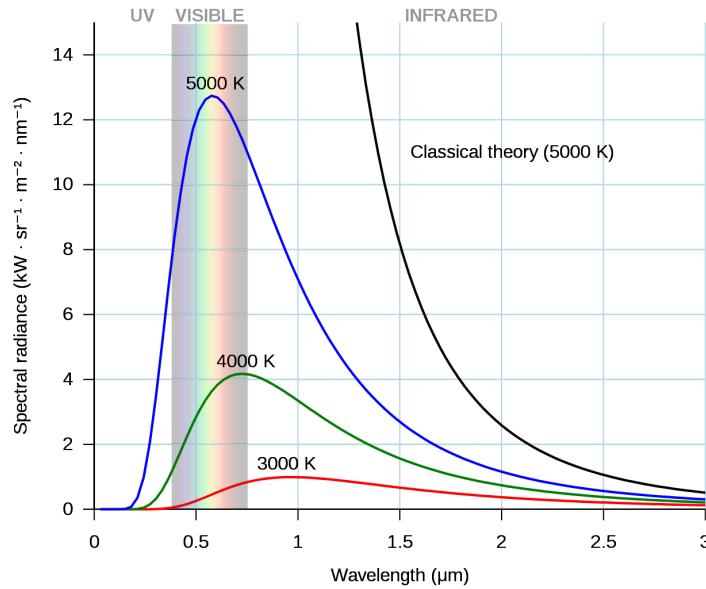
We treat the incandescent bulb as a blackbody since its graphical representation given to us by our broad-range spectrometer resembles a Planck Distribution. The problem we run into is that in order for the peak in the distribution to be where it is, (600 nm range in the spectrum) the bulb's filament would have to be as hot as the surface of a star (at least 5000K), which it can't be. As the voltage is increased, we expect that peak (while it is still inaccurate) to shift to lower wavelengths. We attempt to find an equation for the temperature of the incandescent bulb in terms of voltage. Using the equation $R = R_0[1 + \alpha_{tungsten}(T - T_0)]$. Where R_0 and T_0 are found at room temperature and $\alpha_{tungsten}$ is 0.0045ppm/°C. Solving for T and using $R = I/V$, we find that

$$T = T_0 + \left[\frac{(I/V)}{R_0} - 1 \right] / \alpha_{tungsten},$$

where I is the current (in A) and V is the voltage (in V). The next step would be to plug this equation into the Planck function for T, which gives a final equation of

$$U(\lambda) = \frac{8\pi hc\lambda^{-5}}{\exp\left(\frac{hc}{\lambda k[T_0 + \left(\frac{(I/V)}{R_0} - 1\right)/\alpha_{tungsten}]} - 1\right)},$$

where $U(\lambda)$ is the spectral radiance (intensity) as a function of wavelength, k is $1.38 \cdot 10^{-23} \text{ J/K}$, h is Planck's Constant and c is the speed of light.

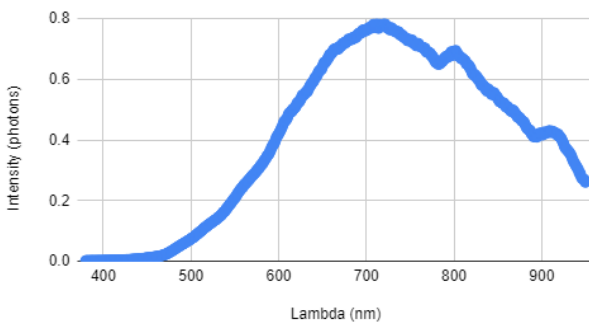


With the function of $U(\lambda)$, we predict that as the voltage increases, the temperature will increase. Subsequently, the peak will occur at lower wavelengths following the condition

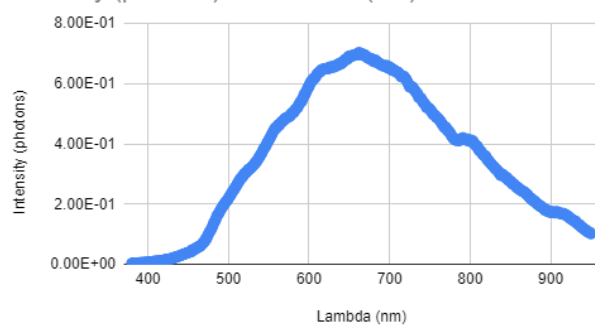
$\lambda_{\text{max}} = 2.89 \cdot 10^{-3} \text{ m/T}$, where λ_{max} is the wavelength associated with the peak intensity and

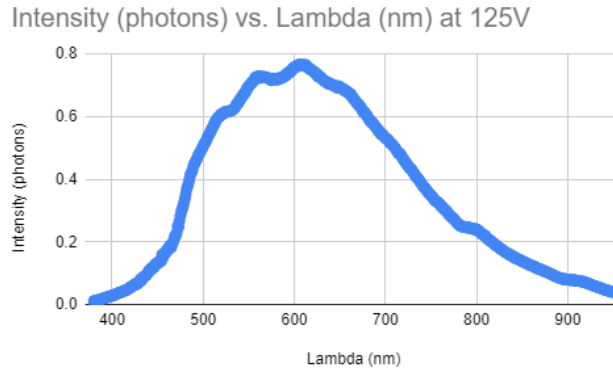
T is the temperature of the blackbody in Kelvin. Even though our spectrometer is inefficient, we can still see this to be true in our data, shown in the figures below, where we only adjusted the voltage.

Intensity (photons) vs. Lambda (nm) at 30V



Intensity (photons) vs. Lambda (nm) at 70V

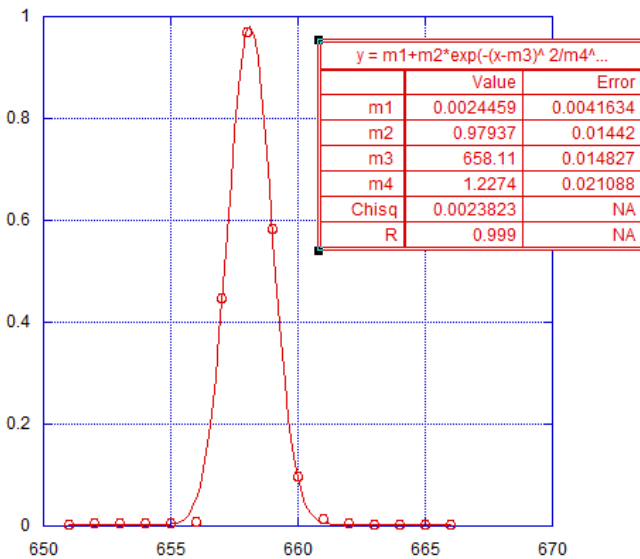




Our distribution shifts in such a way that is consistent with this hypothesis, but we note that the peak seen in the graph is not the peak wavelength, but rather the falloff of our spectrometer in the infrared. In order to fix this we would need an expected value for a set voltage, resistance and temperature, and then we can fit the data and calibrate it similarly to the calibration lamp. We keep this in mind, but we did not have time to actually perform the adjustments.

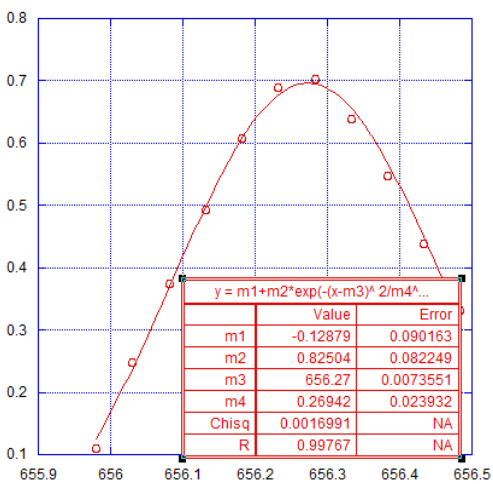
Finally, we understand that hydrogen and helium have emission spectra. This means that when an electron moves down an energy level, it emits a photon with a specific wavelength. The colors of emission lines are consistent with the wavelength of the photon emitted. For hydrogen, we can check if our data is consistent with the theoretical expectation by using the Balmer series for hydrogen: $\frac{1}{\lambda} = R_H \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$, where $n_1 > n_2$ which means that the electron jumps from the n_1 state to the n_2 state when emitting a photon. The photon has a wavelength λ and its energy is equal to the difference in energy between the n -states it starts and ends at (this energy is also the

same as $\frac{hc}{\lambda}$). Starting with hydrogen, we have fit Gaussians at each peak in the spectrum:

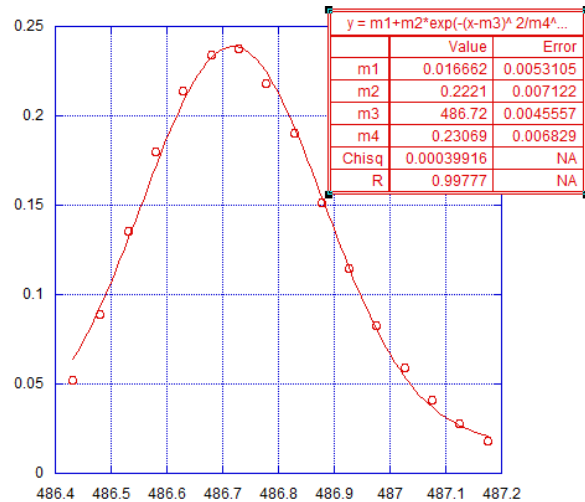


Broadband Hydrogen “658” Peak

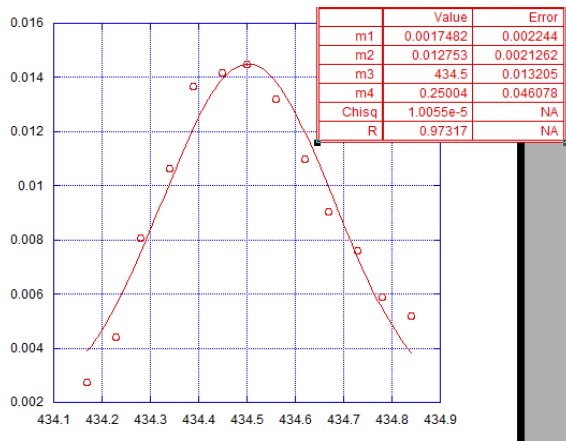
We only include this broadband measurement of a red emission line to show that the broadband spectrometer we use “misses” by a fair amount at higher wavelengths. The Balmer series suggests an emission at around 656 nm for an $n=3$ to $n=2$ jump. This reads 658 nm. Note that the x-axis represents the wavelength (in nm) and the y-axis represents the relative intensity. A higher intensity demonstrates a higher amount of emitted photons. The graphs below are taken with narrow band spectrometers, which were much more accurate to our expected values.



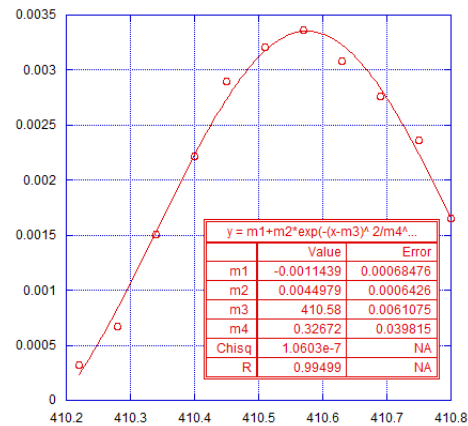
Hydrogen 656 Peak



Hydrogen 486 Peak



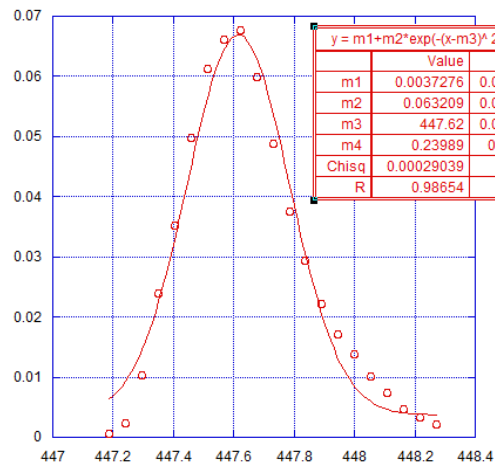
Hydrogen 434 Peak



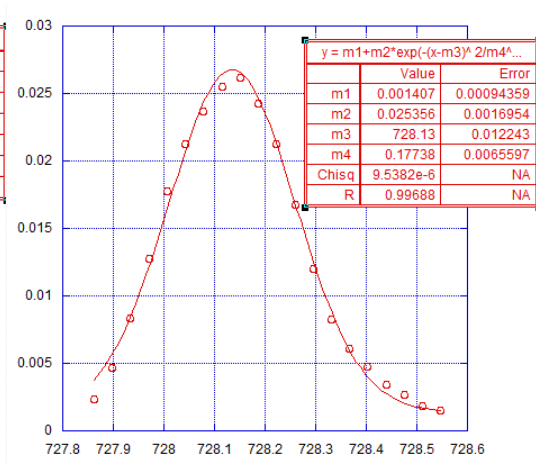
Hydrogen 410 Peak

The Gaussians that we fit here help us to pinpoint wavelengths where emissions occur in the spectrum for hydrogen since we are provided with a “mean” wavelength for each fit. We find 4 emission lines: one in the red range, one in the blue, and two in the indigo-violet range, which is what we expect. The standard deviation we find using the Gaussian fit assists us in determining the line width for each emission line. We estimate that $2 \cdot \sigma$ will give us this value since that is approximately the horizontal width of each emission peak. This is still an estimate, the resolution of the spectrometer is also a factor in determining an accurate line width.

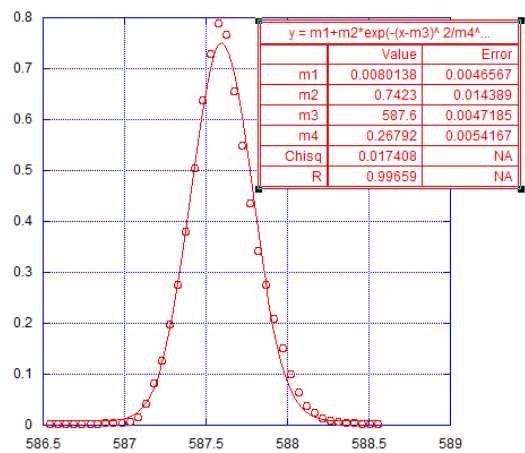
Helium has more than double the number of emission peaks than hydrogen in its spectra since it has more electrons than hydrogen, meaning that more photon emissions will occur. We will only show a few helium Gaussian fits for this reason so that we do not take up space with 10 Gaussian fits.



Helium 447 Peak



Helium 728 Peak



Helium 587 Peak

IV. Conclusions

We conclude that our broad-range spectrometer is very inefficient when detecting higher wavelengths since the incandescent bulb should have its Planck distribution peak in the infrared, but our spectrometer claims it peaks a lot sooner (600-700 nm). This suggests that the filament temperature is much higher than we calculate. The voltage is proportional to filament temperature and increasing the voltage across the incandescent lamp results in an intensity peak in the Planck Distribution at lower wavelengths. The numerical relationship to go along with this observed trend was unattainable in the time allotted for this lab. When taking emission spectra for H and He, our 4 reliable measured emission lines for hydrogen occur at 656.27 nm (linewidth 0.54 nm), 486.72 nm (width of 0.46 nm), 434.5 nm (width of 0.5 nm), and 410.58 nm (width of 0.66 nm). Comparing this to theoretical results from the Balmer series, (656.5 nm, 486.3 nm, 434.2 nm, and 410.3 nm) we applaud our narrow-range spectrometers for reasonably accurate experimental measurements. The highest wavelength emission line in the hydrogen spectrum represents an electron jumping from $n=3$ to $n=2$, the next highest is $n=4$ to $n=2$, the next highest is $n=5$ to $n=2$, and the lowest energy represents a jump from $n=6$ to $n=2$. For helium, we find 10 emission lines across the spectra: 389.31 nm (width of 0.64 nm), 447.62 nm (width of 0.48 nm), 471.82 nm (width of 0.44 nm), 492.79 nm (width of 0.48 nm), 502.24 nm (width of 0.5 nm), 505.44 nm (width of 0.6 nm), 587.6 nm (width of 0.54 nm), 667.97 nm (width of 0.44 nm), 706.59 nm (width of 0.42), 728.13 nm (width of 0.36). To further improve our results for any of these spectroscopy experiments, we should consider blocking out all external light sources that may be picked up by spectrometers. This includes closing blinds, shutting off lights, etc. to minimize any unwanted noise in our detector's readings.